

RoAlgo

Experimental Roblox spring-physics trading algorithm

A rough proof of concept that uses the Roblox engine itself, Luau code plus its native physics solver, to compute a stock-market indicator inside Roblox Studio.

266,268

VERIFIED PHYSICS STEPS

0

CORRUPT STEPS

0

TRADING EDGE FOUND

READ FIRST

This is a research toy, not a trading system. Nothing here places orders, every trade is simulated on historical data, and none of the tests found a trading edge. It is not financial advice.

1 CANDLES IN



MARKET FEATURES

Historical SPY / QQQ five-minute bars become 33 features and eight bounded drive channels: trend, return, range, volume, peers and regime.

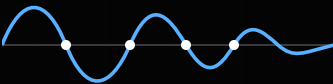
2 SPRINGS MOVE



NATIVE PHYSICS

Each channel pushes real Roblox parts on rails with **VectorForce**. **SpringConstraints** pull them back. Roblox's solver does the motion.

3 MOTION READ



51 MEASUREMENTS

The code reads 24 displacements and 24 velocities per rig and adds three bank energies. The springs act as a memory of recent pushes.

4 DECISION OUT

BUYSELL

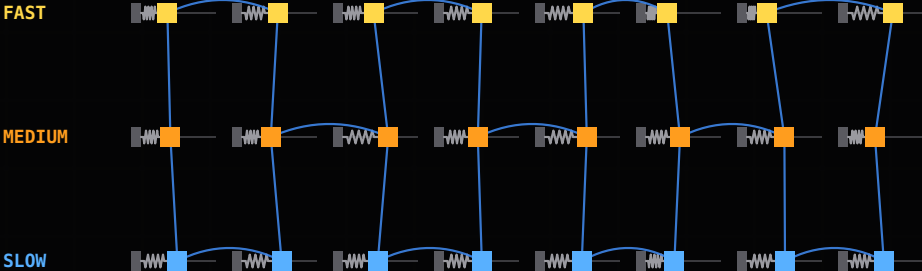
CLOSEHOLD

SIMULATED LEDGER

A trained readout, or in v3 the spring states directly, produces BUY, SELL, CLOSE or HOLD. Trades are scored after declared costs.

THE RIG

coupled · 24 native nodes · 37 coupling springs



■ Fast row: immediate returns, range, activity ■ Medium: trend, peers, hourly context

■ Slow: daily returns, regime

Each node is an unanchored **Part** on a **PrismaticConstraint** rail with a restoring spring. A second, independent rig has the same 24 nodes without the blue coupling springs, so the two can be compared. Every five-minute bar is held for 12 physics steps of 1/60 s.

ENGINE FINDING

verified stepping

In Studio 0.741 Edit mode, **WorldRoot:StepPhysics** lands a frame late and drops about one request in eight. So every step is verified:

- Two large "metronome" parts ride along in every call.
- Only one request is outstanding at a time.
- Each step is confirmed to be exactly +1 dt.
- Dropped requests are re-issued after three frames.

Details: docs/stepping-capability.md

WHAT'S IN THE REPOSITORY

four iterations, early October 2026

lab/	v0, forecast lab. A 24-node spring and rail reservoir with ridge regression, a Markov baseline and a preregistered, sealed test period.
indicator/	v1, first indicator. Fast, medium and slow spring filters giving long/flat BUY, HOLD and SELL in a Studio dashboard.
indicator-v2/	v2, learned mechanics. Coupled and independent rigs, a numerical twin, a regime model, a long/short ledger and replay.
indicator-v3/	v3, Market Lab and Spring Indicator. Sector inputs, 24 and 48-node rigs, and decisions read directly from spring states.
tools/ docs/	Node orchestration, packaging and validation; design plans, the lab runbook and the stepping notes.

The physics computes.
It does not beat doing nothing.

Decisions measurably depend on the spring states, but no model beat a "no change" forecast or "no trade" by a meaningful margin.

RESULTS					simulated under each experiment's declared cost model, not broker fills
EXPERIMENT	PERIOD	PHYSICS	COMPARISON	WHAT IT SHOWED	
v0 lab 8-hour forecasts, preregistered	2025-01-01 to 2026-10-02	+0.0040 nRMSE, higher is worse	CI -0.0001...+0.0093	Physics did not help. Every model was within about 1% of a "no change" forecast.	
v1 indicator	5 sessions, Dec 2024	SPY -0.69%	-0.95%	Lost money after costs. The numerical control lost more.	
		QQQ -0.55%	-1.50%		
v2 learned readout	5 sessions, Dec 2024	-0.73%	raw +0.64%	Not a meaningful comparison: a later check found the readouts were far worse than a zero forecast.	
v3 repaired readout	same saved v2 states	no_trade	zero forecast	Every model still loses to the zero forecast on validation.	
Spring Indicator	10 sessions, Dec 17-31 2024	-2.58%	EMA -2.25%	Already-inspected sessions. Independent springs -1.74%, numerical -1.63%.	
			no-trade 0%		
Spring Indicator 34,134 bars replayed in Studio	2025-01-02 to 2026-10-02	-56.1%	EMA -67.1%	The predeclared edge criterion failed. Independent -57.4%, numerical -54.1%.	
			no-trade 0%		

WHAT YOU NEED

market data is not included

Roblox Studio

Developed on 0.741. Edit mode, local plugins, HTTP requests allowed.

Luau CLI 0.741

For the offline tests. Put **luau** on your PATH or set **LUAU_EXE**.

Node.js 24

No npm dependencies. Serves cached data to Studio over localhost.

Alpaca API key

Your own key for the read-only market-data API. Downloaded bars must not be shared.

1 · GET THE DATA

from the repository root

```
node download.mjs <credentials-file> data/alpaca-2021-10-04_2026-10-02
node prepare.mjs data/alpaca-2021-10-04_2026-10-02
node download-supplement.mjs <credentials-file> \
  data/alpaca-supplement-2016-01-01_2026-10-02
node prepare-supplement.mjs data/alpaca-supplement-2016-01-01_2026-10-02
node audit-supplement.mjs data/alpaca-supplement-2016-01-01_2026-10-02
node indicator-v3/fetch-sector-context.mjs <credentials-file>
```

Keep the credentials file outside the repository. The folder names and date ranges are fixed in the scripts.

2 · RUN THE LATEST VERSION (V3)

```
node indicator-v3/package.mjs # builds the plugin
node indicator-v3/bridge.mjs # bridge on 127.0.0.1:4762
```

- Copy **indicator-v3/build/RoAlgoMarketLab.plugin.rbmxm** to **%LOCALAPPDATA%/Roblox/Plugins/RoAlgoMarketLab.rbmxm**, or run **Install-In-Studio.luau** from the Command Bar.
- Open an unpublished place in Edit mode. The plugin button toggles the dashboard.
- On Windows, **Start-RoAlgo.ps1** starts the bridge for you.

3 · RUN THE TESTS

no Studio or market data needed

```
npm test # 141 Node tests
npm run test:luau # 37 Luau spec files
```

npm run test:all also runs tests that check the original private data by exact hash, so those only pass with the original files.

GROUND RULES

- Do not publish market data.** Alpaca does not allow redistribution, so **data/**, **results/** and **runs/** are git-ignored.
- Use a dedicated place file.** The plugins check **game.PlaceId** and start in any unpublished place (PlaceId 0).
- The Spring Indicator cannot be rerun as-is.** It is pinned to the original saved capture, which is not included.
- Treat it as experimental code.** Most of it was written by AI coding agents under the author's direction.